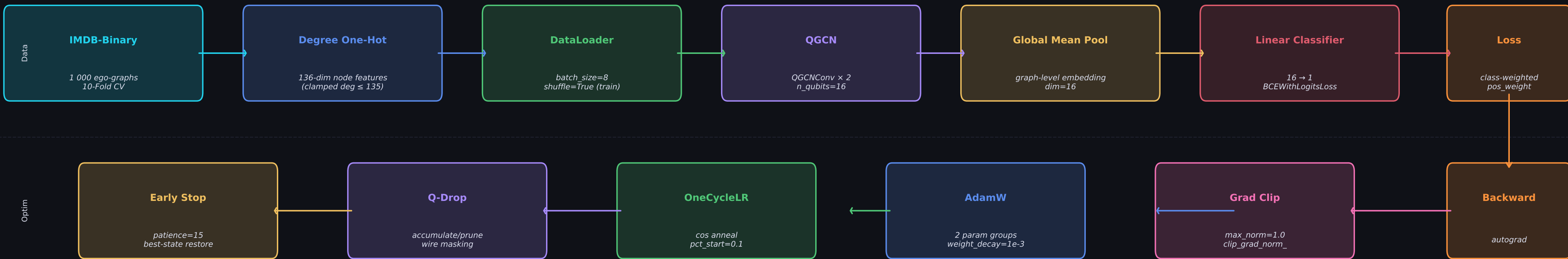


Q-Drop Quantum GCN — Full Training Pipeline & Learning Rate Schedule

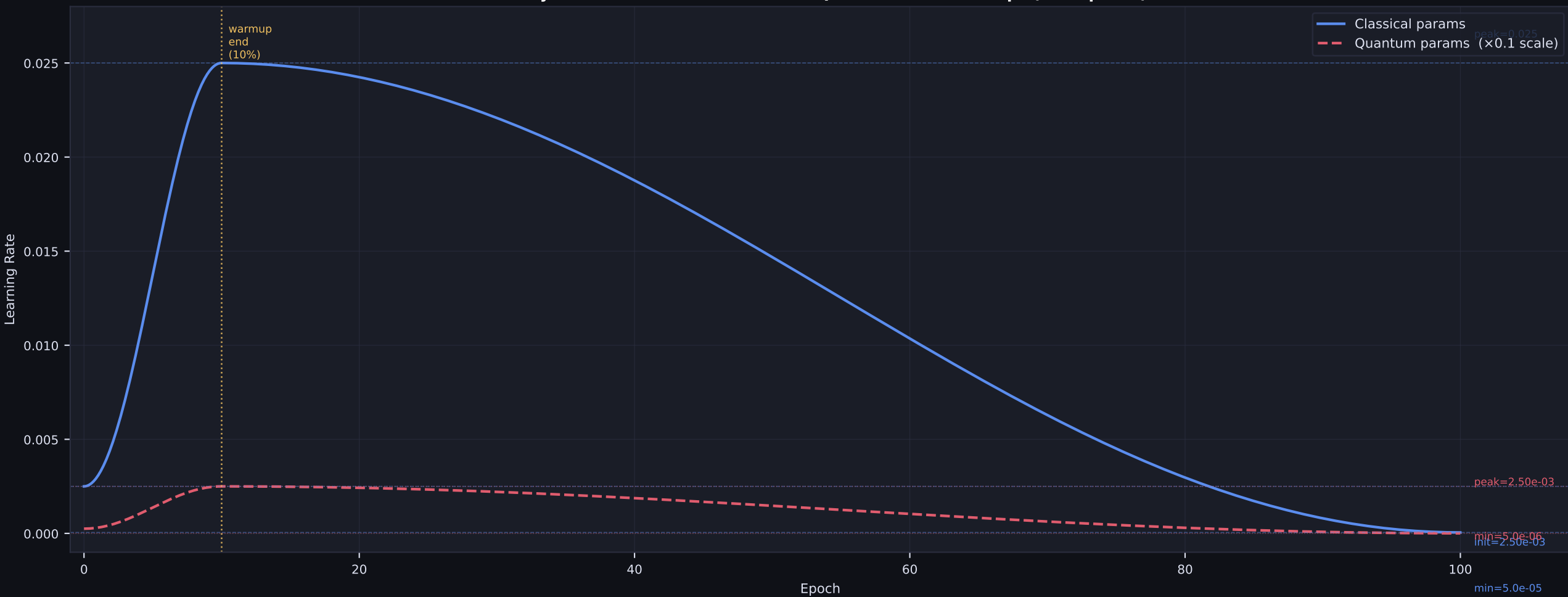
Q-Drop Training Pipeline — IMDB-Binary

FORWARD PASS

BACKWARD PASS



OneCycleLR Schedule — Classical vs Quantum Param Groups (100 Epochs)



LR Calculation — Key Values

OneCycleLR Parameters

```
base_lr      = 5e-03
pct_start    = 0.1
div_factor   = 10
final_div_factor = 50
anneal_strategy = 'cos'
```

Classical group

```
max_lr = base_lr × 5 = 0.0250
init_lr = max_lr ÷ 10 = 0.0025
min_lr = init_lr ÷ 50 = 5.0e-05
```

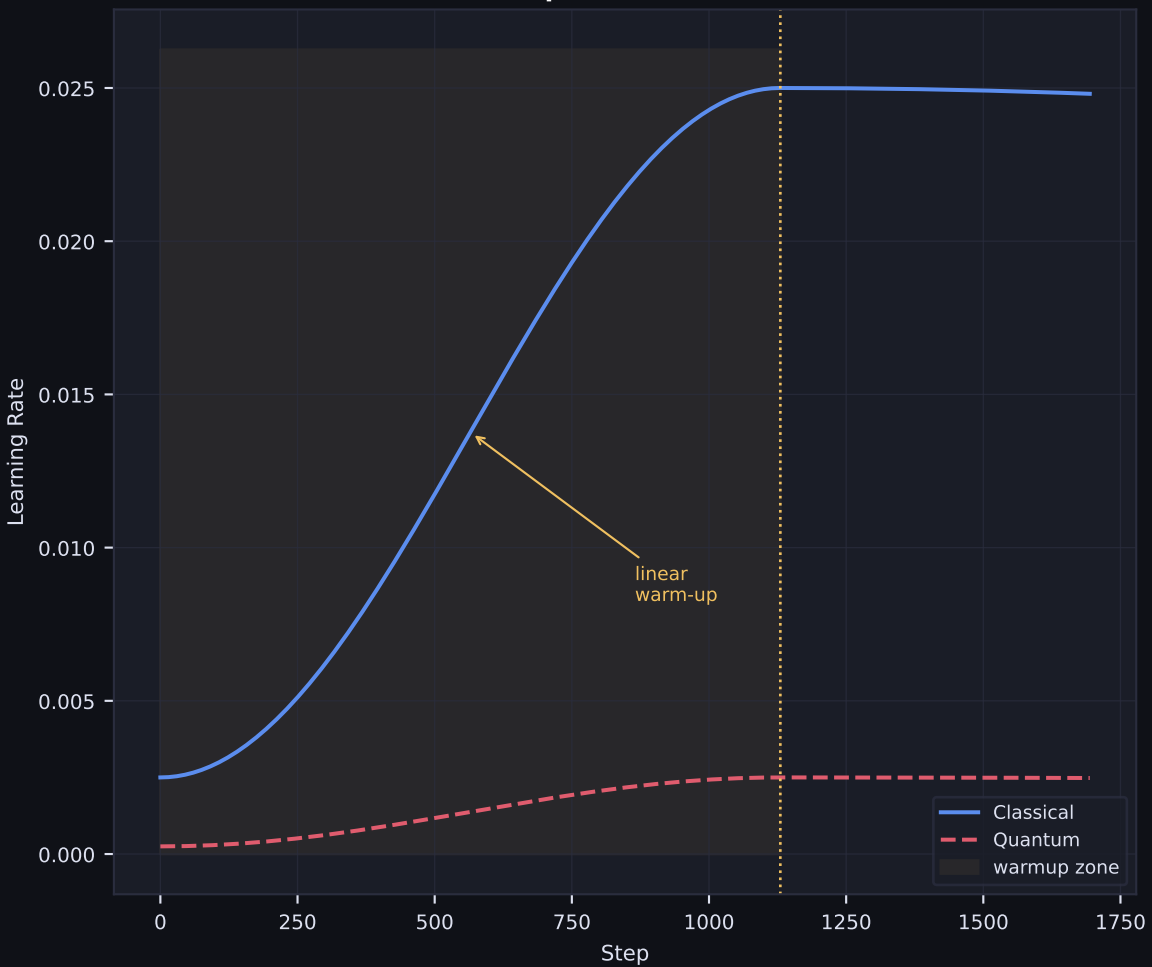
Quantum group

```
scale = quantum_lr_scale = 0.1
max_lr = 0.0250 × 0.1 = 2.5000e-03
init_lr = 2.50e-03 ÷ 10 = 2.50e-04
min_lr = 2.50e-04 ÷ 50 = 5.00e-06
```

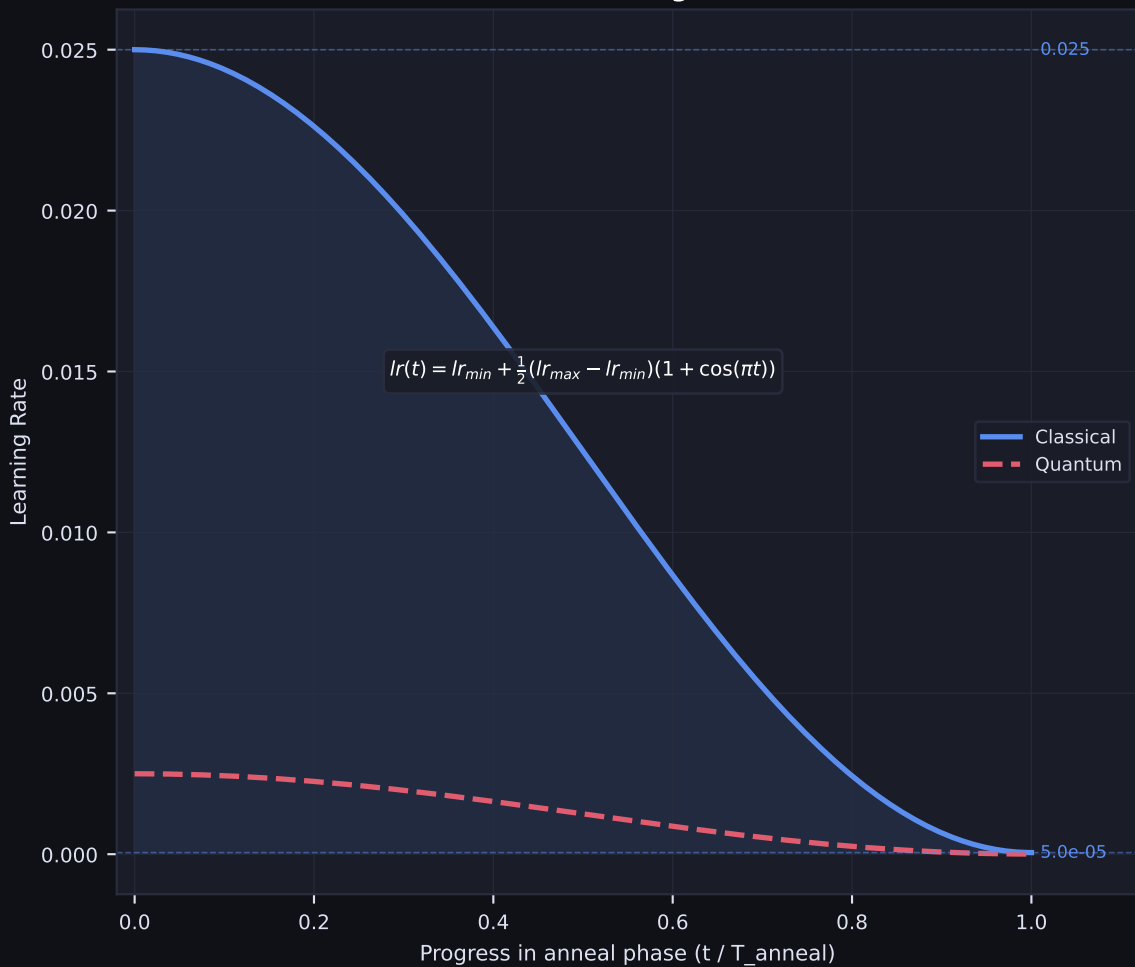
total_steps

```
= epochs × steps_per_epoch
= 100 × 113 = 11,300
warmup = 1,130 steps (10%)
anneal = 10,170 steps (90%)
```

Warmup Phase (0 → 10%)



Cosine Annealing Formula



AdamW Update Rule

Step t (per param group):

```
g_t = ∇L (+ grad_clip ≤ 1.0)
m_t = β₁ · m_{t-1} + (1 - β₁) · g_t
v_t = β₂ · v_{t-1} + (1 - β₂) · g_t²

m_hat_t = m_t / (1 - β₁^t) [bias corr]
v_hat_t = v_t / (1 - β₂^t) [bias corr]

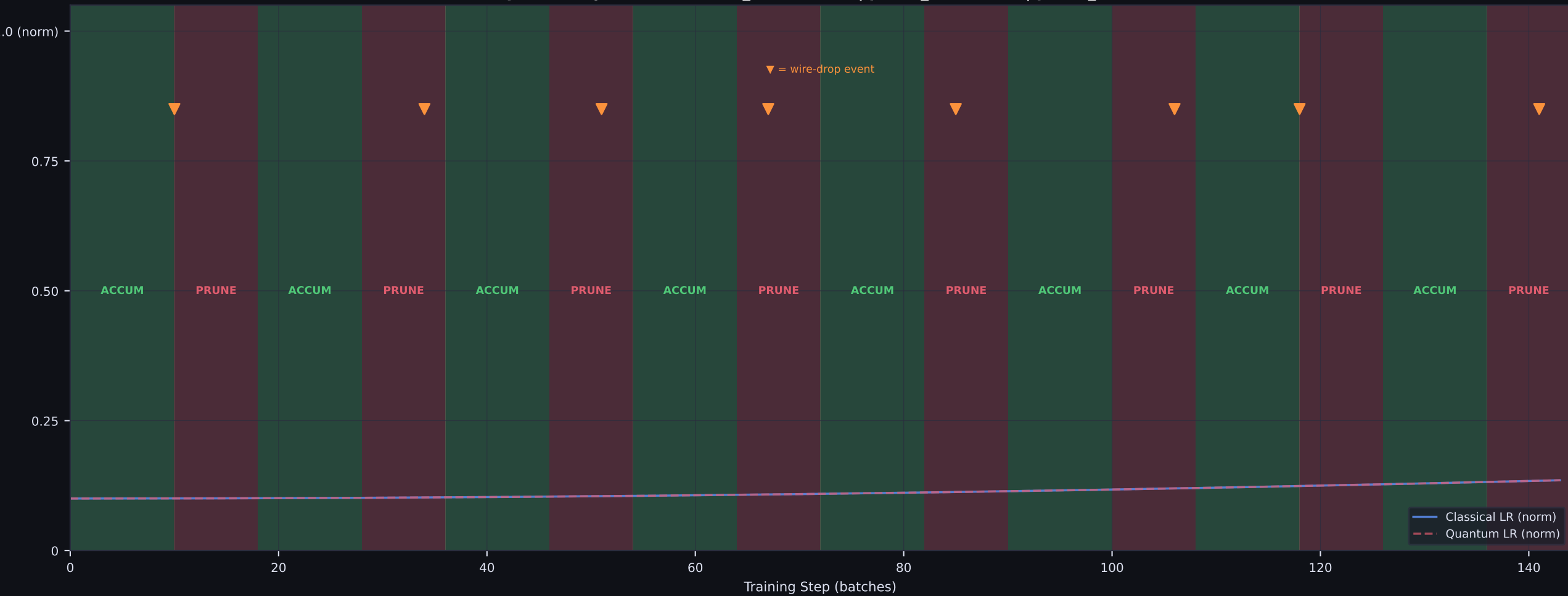
θ_t = θ_{t-1}
      - lr(t) · m_hat_t / (√v_hat_t + ε)
      - lr(t) · λ · θ_{t-1}
      + weight_decay=1e-3
```

Defaults: β₁=0.9 β₂=0.999 ε=1e-8

Two lr values from OneCycleLR:

```
Classical: 0.0025 → 0.0250 → 5.0e-05
Quantum:   2.50e-04 → 2.50e-03 → 5.0e-06
```

Q-Drop Phase Cycle (accumulate_window=10 | prune_window=8 | prune_ratio=0.8)



10-Fold CV + Early Stopping

